

USER QUERY & CANDIDATE APIS

[Kiểm tra phạt nguội cho xe máy biển số 59P1-12345 tại TP. Hồ Chí Minh]

[EN: Check traffic fines for motorbike 59P1-12345 in HCMC]

Candidate Tools: [tra_cuu_phat_nguoi, dat_ve_xe_khach, thanh_toan_tien_dien,...]

Method 1: End-to-End Generative SLM

Qwen3.5 (2B/4B) + QLoRA SFT (Response-only loss)

Autoregressive Decoding (Native XML tags)

Direct JSON Tool Call Output

Method 2: Bi-Encoder + Cross-Encoder

Dense Retrieval (BGE-M3 + CachedMNRL) → Top-3 Candidates

Abstention & Threshold Gate ($\tau=0.35$, $\delta=0.21$)

Hierarchical Cross-Encoder (XLM-R)

 has_value

 Span Extraction

 Enum/Bool

Value Normalizer & Schema Validation → Deterministic JSON